One genome in, and what each stage hands the next

NC_045512.2 · identified through its own curated entries · every count is read from runs/onset_chain.json, written by python -m hpc_pilots chain

INPUT · one pathogen genome accession

NC_045512.2 · 29,903 nt · 1 segment
Nothing else: no target name, no shortlist, no structure.

genome_accession

ARM 1
the onset co-scientist

five stages, 221 s end to end, live against public endpoints

1 · intake.rank_targets

NCBI · UniProt · 11.21 s
in: the accession, and nothing else

out
28 ORFs called in six frames, 12 ranked
9 identified against 17,529 reviewed entries
3 not identified, and reported as such

target_name · protein_accession · residue_ranges · protein_sequence
— all four facts the other nine tools are waiting on, from this one call

2 · intake.decompose + rank_score

UniProt · 111.26 s
in: 9 identified candidates

out
21 mature chains, each with a score
score = 3·conservation + 2·exposed + 1.5·essential
+ 0.5·min(1, structures/50) + 0.8·druggable
every term is a stated weight; nothing here is fitted

one accession and one residue range per chain — 21 of 21 go on,
and 21 of 21 can also supply their own sequence to stage 4

3 · geometry.surface_markers

PDBe · RCSB · UniProt · 96.78 s
in: 21 chains, each as accession + residue range

out
21 rows for 21 chains: 20 measured, 1 refused with a reason
largest exposed hydrophobic patch, median of 5 structures
gate: 2 yes · 2 borderline · 16 no

a patch median and a patch range per chain

4 · structures.folded

ESM Atlas · 2.1 s
in: one amino-acid sequence — from stage 2, not stage 3

out
21 of 21 chains could supply their own sequence
folded 3C-like proteinase nsp5, 306 residues
mean pLDDT 62.57, 46.4% at or above 70
— the predictor's confidence in itself, not accuracy

THE ARROW THAT IS NOT THERE

Stage 5 reads nothing stages 1 to 4 measured. `library_spec` takes an initiator, a length range and a set of elongator classes — no patch area, no accession, no residue range. The library is the same library whatever the surface stage found, so "a focused library over that surface" states an intent and not a data path.

5 · chemistry.library_spec

local tables · 0.0 s
in: nothing from upstream

out
ClAc-D-Tyr initiator, 4–8 randomised positions
over 25 building blocks, C closes the ring by thioether
158,945,703,125 distinct sequences allowed

how many positions the library randomises (4–8)
— the only number that crosses from one arm to the other

ARM 2
will a model fitted on the first round be worth acting on?

answerable at design time, and the strongest result here

6 · the go / no-go, before any assay

in: how many positions the library varies, and whether the candidates vary at the same positions as the members already measured. No assay. No structure. No target.

out
candidates share the measured positions → +0.173 to +0.321
candidates share none of them → +0.010 to +0.033
gain over random from the same batch, on 128 measurements

a decision, before anything is synthesised:
build this library, or redesign it so the positions overlap

WHAT WOULD CLOSE THE LOOP, AND IS NOT WIRED EITHER

Measured members come back from a campaign as the first round a model is fitted on, and the go/no-go runs on the design that produced them. Eight registry fields are that interface, named exactly, and all eight are empty for all 29 blocks: 232 cells of 232.